

Magic Mirror: Design of a Computer Vision-Based Biomechanical Interface

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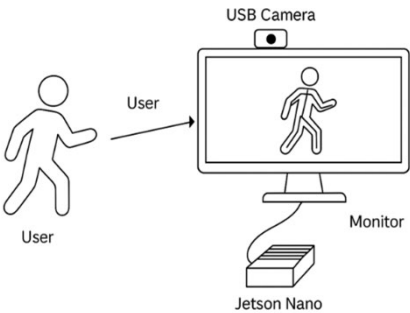
BACKGROUND

- Problem:** Traditional biomechanical analysis systems (e.g., Vicon, OptiTrack) are expensive, lab-based, and require expert setup.
- Need:** There is a growing demand for real-time, portable, and user-friendly systems in physical therapy, sports science, and home fitness.
- Solution:** Magic Mirror offers a low-cost, edge AI-based interface for human motion tracking using computer vision.
- Built on the NVIDIA Jetson Nano, it leverages MediaPipe BlazePose, OpenCV, and a Flask-based UI to capture, analyze, and display joint movement data in real time.
- The system enables users to receive live feedback on form, reps, joint angles, and performance metrics—without the need for a clinician or lab environment.

Magic Mirror bridges the gap between high-end lab technology and everyday accessibility, promoting smart movement for everyone.

OBJECTIVES

- Develop a real-time movement tracking system using computer vision and edge AI.
- Capture skeletal keypoints during exercises using MediaPipe BlazePose.
- Compute biomechanical metrics such as joint angles, range of motion (ROM), repetition count, and angular velocity.
- Provide users with live, on-screen feedback during movements like squats, lunges, sit-to-stands, leg raises, and jumps.
- Enable a user-friendly interface using Flask, compatible with both local and web-based access.
- Ensure the system is low-cost, scalable, and portable for use in home fitness, physical therapy, and athletic settings.



Physical Setup of Magic Mirror

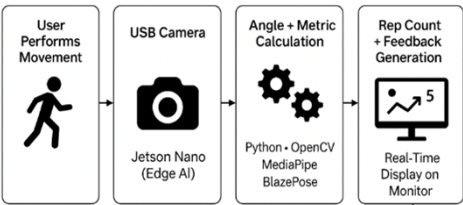
DESIGN PROCESS

Materials Used

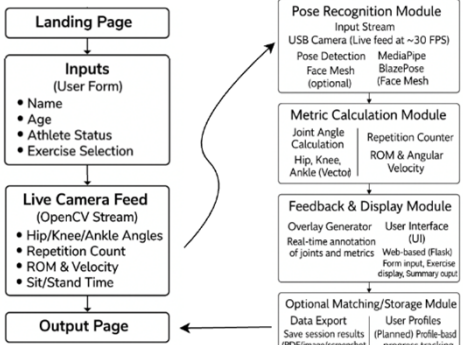
Component	Purpose
Jetson Nano (4GB)	Edge AI computation for real-time analysis
USB Camera	Captures human movement
HDMI Monitor	Real-time visual feedback display
Mouse + Keyboard	Navigation and UI interaction
Python, OpenCV	Image processing & camera interfacing
MediaPipe BlazePose	Pose detection and landmark tracking
Flask + HTML/CSS/JS	Web interface for user interaction

Programming Function

- System Initialization:** The Jetson Nano powers up, connects to camera and monitor, and launches a Flask server to manage the UI.
- Data Capture:** Video streamed via OpenCV and processed by BlazePose to extract 33 pose landmarks in real time.
- Pose Processing:** Hip, knee, and ankle angles are computed using vector math. Metrics like ROM, angular velocity, rep count, and timing are calculated.
- Feedback Output:** The monitor displays a skeleton overlay with live annotations of joint angles and movement metrics.
- Web Interaction:** Users input their details and select an exercise through the Flask UI. The system runs the matching script and shows session results.



Data flow in Magic Mirror system



Software Architecture Diagram

RESULTS

System Capabilities

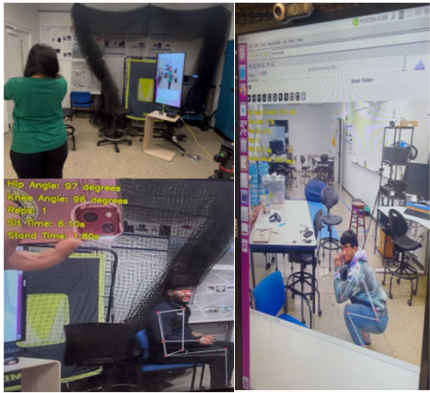
- Exercises tested: Squats, Lunges, Leg Raises, Sit-to-Stand, Jumps, and Face Mesh
- Average frame rate: ~30 FPS on Jetson Nano
- Detected 33 pose landmarks and 468 face landmarks (using MediaPipe Face Mesh)

Pose Estimation Outputs

Exercise Type	Key Metrics Tracked
Squats	Hip angle, knee angle, ROM, rep count, angular velocity
Lunges	Left vs right leg angle, rep count, balance tracking
Leg Raises	Peak leg height, left/right angle, rep symmetry
Sit-to-Stand	Sit time, stand time, hip/knee transition angle
Jump	Flight time, impact angle, joint compression
Face Mesh	Eye open/close detection, blink tracking, face orientation detection

Performance Summary

- Compliance:** IRB Approval obtained for participant testing
- Trial Size:** 14 gym enthusiasts + 10 student participants
- Repetition Accuracy:** >95% compared with manual count
- Face Mesh Blink Accuracy:** >97%
- Average Time Deviation (Sit/Stand):** ±0.3s compared to stopwatch
- Latency:** ~0.2 seconds from motion to on-screen feedback
- User Satisfaction:** Average usability rating of 4.6/5



Demonstration of the Magic Mirror system in use across different environments.

Exercise	Key Metrics	Avg Accuracy (%)	Latency (s)	Notes
Squats	Hip/Knee Angle, ROM, Reps, Angular Velocity	96	0.2	Smooth tracking and repetition detection
Lunges	Leg Angle (L/R), Stability, Reps	94	0.2	Side tracking stable, balance affects precision
Leg Raises	Leg Height, Side Angle, Reps	93	0.2	Height detection accurate in standing posture
Sit-to-Stand	Sit Time, Stand Time, Transition Angle	95	0.2	Consistent timing detection
Jump	Flight Time, Compression Angle	92	0.25	Slight frame delay in capture
Face Mesh	Blink Count, Eye Status, Face Orientation	97	0.15	Real-time blink/eye status highly stable

CONCLUSION

The Magic Mirror system successfully delivers real-time biomechanical analysis using a low-cost, embedded AI setup. With high accuracy in joint angle detection, repetition counting, and facial expression monitoring, it effectively supports lower extremity analysis and basic upper body tracking.

Through the use of Jetson Nano, MediaPipe BlazePose, and Face Mesh, the system provides valuable live feedback for users performing exercises such as squats, lunges, leg raises, and sit-to-stand transitions. Its responsiveness and modular design enable applications across physical rehabilitation, athletic training, and remote wellness monitoring.

Who can benefit from it?

- Physical therapists for affordable home rehab monitoring
- Athletes/trainers needing instant form feedback
- Rehab centers for post-injury recovery tracking
- Fitness beginners seeking guided real-time correction

FUTURE WORK

- User Profiles:** Enable session tracking with visual progress summaries.
- Heart Rate Sync:** Integrate Bluetooth monitors for biometric feedback.
- Remote Monitoring:** Allow therapists and coaches to access user data via the cloud.
- AI-Based Scoring:** Classify posture as ideal, needs improvement, or incorrect.
- Inclusive Design:** Add support for colorblind users, audio cues, and larger text.
- Motivational Features:** Introduce badges, goals, and achievements to boost engagement.
- Advanced Gait Analysis:** Detect symmetry issues and fall-risk through walking assessment.

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